



Web Development & UX/UI design

Sisters of Code Employment Skills Program

Funded by Liechtenstein Development Service

Week 1

Content

1. Basic introduction to computer, files and browser
2. How internet works
3. Introduction to website and its benefits
4. Introduction to Web Development & Design
5. Understand website development
6. Set up local environment

Basic Computer

For a computer to operate, we need **Hardware** and **Software**.

- The **hardware** (CPU, RAM, Keyboard, motherboard, Monitor screen, mouse, trackpad, etc.) is the body.
- The **OS** is the brain that tells everything how to work together.
E.g. Microsoft Windows 7 / 8 / 10, Mac OS Ventura 13.7
- The **applications** are tools you use — and the OS helps them run.
E.g. Google Chrome, Microsoft Word, Telegram

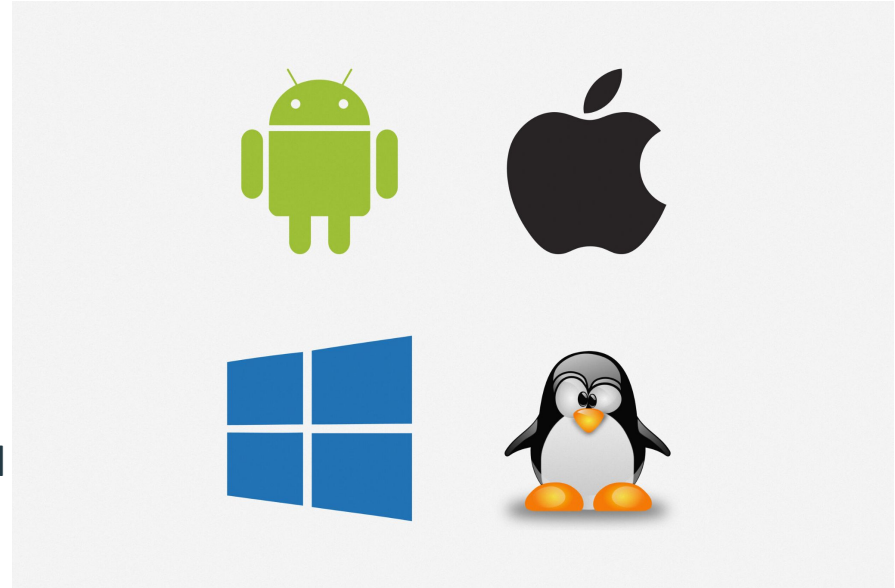


Basic Computer

What is an Operating System?

An Operating System is the main software that manages all the hardware and software resources on a computer or device. It acts like a bridge between you (the user) and the computer's hardware.

There are different operating systems created by different IT companies.



Can you name these OS?

Basic Computer

Main Functions of an OS

- Manages files and folders (File System)
- Runs applications
- Communicate with hardware devices
 - (keyboard, screen, printer, etc.)
- Handles multitasking
- Manages memory (RAM) and processes

Examples of Operating Systems:

- Windows (Microsoft)
- macOS (Apple)
- Linux (Open-source, used on servers and desktops)
- Android (Google, used on phones/tablets)
- iOS (Apple, used on iPhones/iPads)



Basic Computer File Management

What is File Management?

- The process of organizing and storing files on a computer.
- Helps in easy retrieval and prevents data loss.

Key Concepts:

- Saving Files: Choosing appropriate file names and locations.
- Folders & Directories: Organizing files into categories.
- File Extensions: (.docx, .pdf, .jpg, .html) – Understanding different file types.
- Browser Basics: Navigating, bookmarking, and clearing cache.

Files formats and sizes

Common Text Formats:

- **.txt**: Plain text, very small file size, limited formatting. Good for simple information.
- **.pdf**: Portable Document Format, preserves formatting, good for consistent viewing/printing (e.g., terms, resumes). Can be larger.
- **.doc/.docx**: Microsoft Word format, rich formatting. Can be larger, best for editing before web.

Other Important Formats:

- **.html**: Web page structure and content.
- **.css**: Website visual styling.
- **.js**: Website interactivity.

Image Formats (Key for Websites!):

- **.jpg/.jpeg**: Good for photos, lossy compression (smaller size, some quality loss). Optimize for web.
- **.png**: Supports transparency, lossless (better quality, potentially larger). Ideal for logos, clear backgrounds.
- **.gif**: Animation & transparency, limited colors. Use sparingly.
- **.webp/avif**: Modern, excellent compression & quality, transparency & animation. Recommended for web.

File Size Matters for Websites!

- Smaller files = faster loading = better user experience & SEO.
- Optimize images! Use the right format & compression.
- Keep download sizes reasonable.

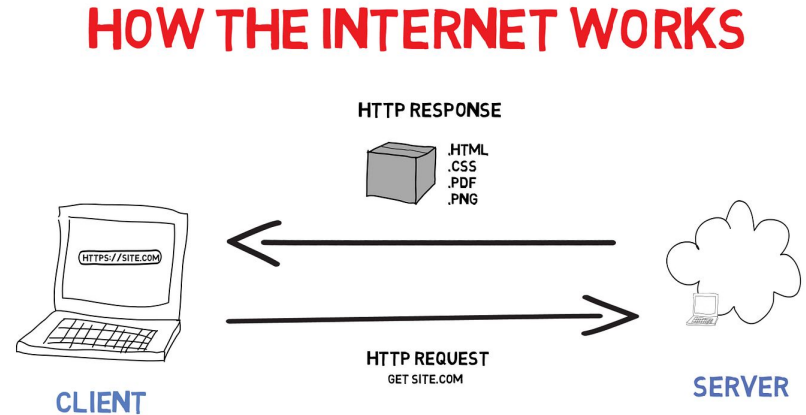
How the Internet Works

How Do We Access Websites?

- Client (Your Computer/Phone) sends a request to a web server.
- Server processes the request and sends back a response.
- Browser renders the webpage for the user.

Key Terms:

- Request: Asking for a webpage.
- Response: Receiving the requested data.
- IP Address: A unique number for each device on the inter



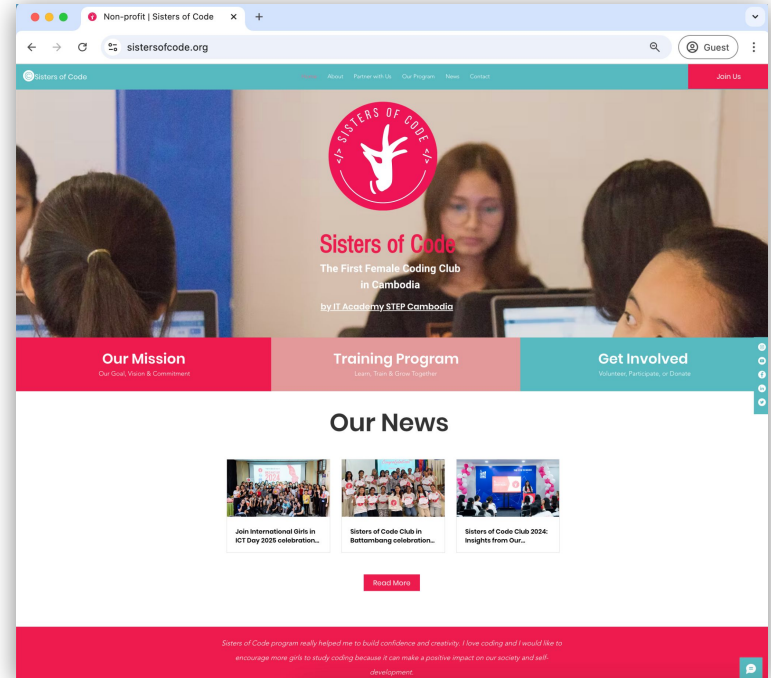
Introduction to Websites

What is a Website?

- A collection of web pages accessible via the internet.
- Can be for business, education, entertainment, etc.

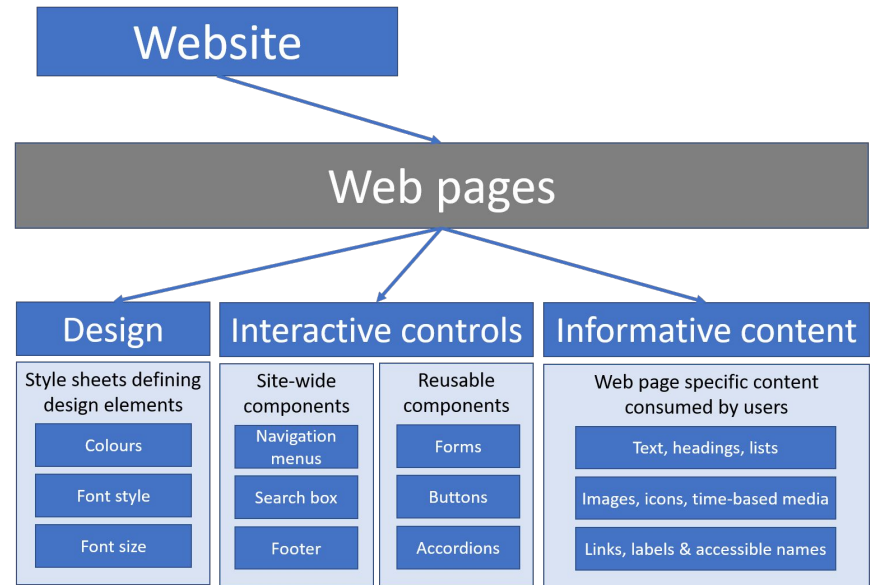
Why Are Websites Important?

- **Visibility:** Allows businesses and individuals to reach a global audience.
- **Communication:** Helps in sharing information and connecting with others.
- **Marketing & Sales:** E-commerce, online branding, and business growth.
- **Content:** Make it possible to share content or information in many forms and manage access to those resources



Website Fundamentals

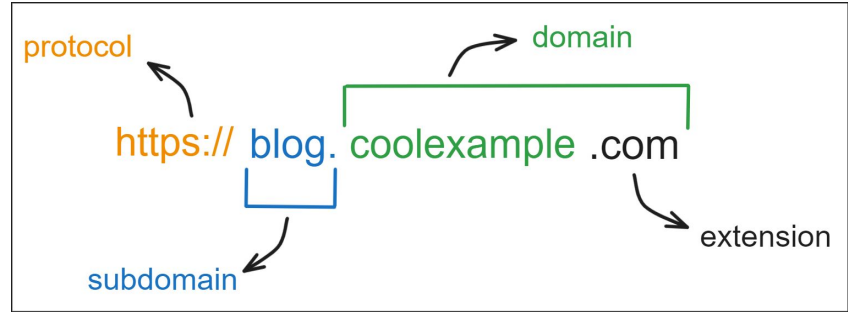
- **Website:** A collection of linked web pages.
 - <https://www.sistersofcode.org/>
 - <https://www.sistersofcode.org/about>
 - <https://www.sistersofcode.org/partner-with-us>
 - <https://www.sistersofcode.org/our-program>
- **Port:** A communication endpoint
 - 80 for HTTP
 - 443 for HTTPS
 - 21 for FTP
 - 22 for SSH
 - Custom port:
 - <http://website.com:9090/page/1>
 - <http://localhost:8000/example>



Website Fundamentals Cont'

- **Domain Name:** The website's address

- **www.google.com**
- **www.sistersofcode.org**
- **www.souladvisor.com**
- **news.sabay.com.kh**
- **intocambodia.com**
- **monoria.biz**



- **Host:** The provider where a website's files are stored

- **Web Host** (Website Hosting Provider): A company that provides space on a server to store and serve website files. Examples: Bluehost, Hostinger, SiteGround, AWS, Google Cloud.
- **Local Host** (Local Development Environment): Refers to your own computer acting as a server for web development. Examples: Tools like XAMPP, WAMP, or localhost (127.0.0.1)
- **Server Host** (Remote Hosting for Applications): A remote server where web applications, databases, and APIs are hosted. Examples: DigitalOcean, Linode, AWS EC2, Heroku.

Website Fundamentals Cont'

- **IP Address:** A unique number assigned to each device connected to the internet.

- IPV4:

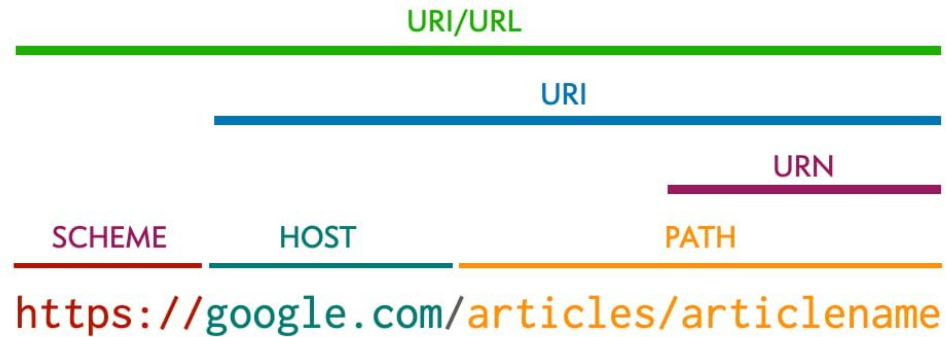
- 0.0.0.0
- 192.168.0.1

- IPV6:

- ff06:0:0:0:0:0:c3
- 38:f9:d3:60:37:a1

- **URL vs. URI:**

- URL (Uniform Resource Locator)
- URI (Uniform Resource Identifier)
- URN (Uniform Resource Name)



How Websites Are Rendered

Why Are Different Languages Needed?

=> They have their own jobs!

- **HTML** (HyperText Markup Language): The structure and content of a webpage
 - Creates a button.
 - Create text e.g. headings, paragraphs, list, images
- **CSS** (Cascading Style Sheets): The design and layout.
 - Styles the button with colors and shapes.
- **JavaScript** (JS): The interactivity and dynamic content.
 - When clicked, it shows an alert
 - Slideshow
 - Modal popup
 - Fetch data or change content without reload page

How Hosting Works

What is Web Hosting?

- A service that allows individuals and businesses to post a website online.
- Websites need hosting to be accessible to users.
- In short, a storage service!

Types of Hosting:

- Shared Hosting: Multiple websites share the same server.
- Dedicated Hosting: A full server for one website.
- VPS (Virtual Private Server): A dedicated portion of a server.
- Cloud Hosting: Scalable hosting across multiple servers.

Introduction to Web Development & Design

The Website Creation Process:

- **Planning:** Defining purpose, audience, and content.
- **Designing:** Wireframes, color schemes, and layouts.
- **Development:** Writing code using HTML, CSS, and JavaScript.
- **Testing & Deployment:** Debugging and making the site live.
- **Maintenance:** Regular updates and improvements.

Skills Required:

- HTML, CSS, JavaScript
- Problem-solving & debugging
- Creativity & user experience (UX) design

Tools Used in Web Development:

- Code Editors: **VS Code** / Sublime Text / PHPStorm
- Version Control: **Git**, **GitHub** / Gitlab
- Design Tools: Figma, **Framer**, **Canva**

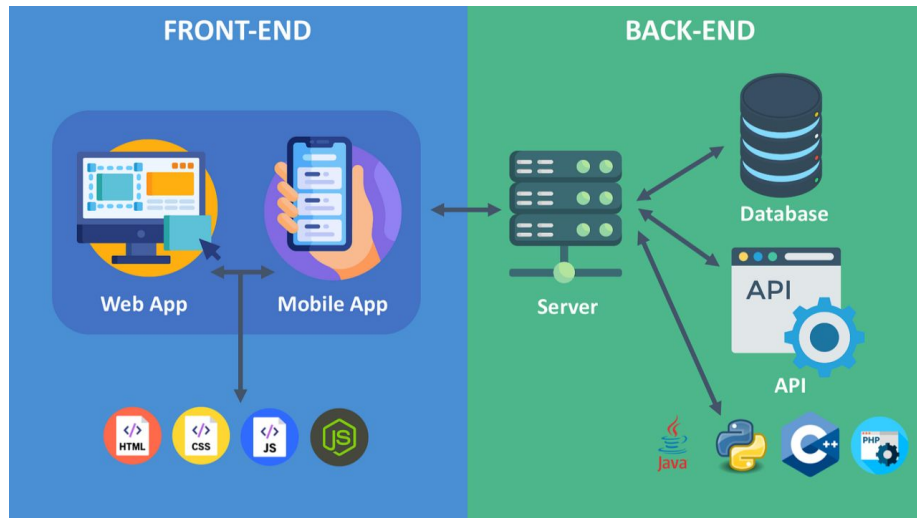
Understanding Website Development

Static vs. Dynamic Websites

- ❖ Static Website: Fixed content, loads the same for every user (e.g., a personal portfolio).
- ❖ Dynamic Website: Changes based on user interaction (e.g., Facebook, Amazon).

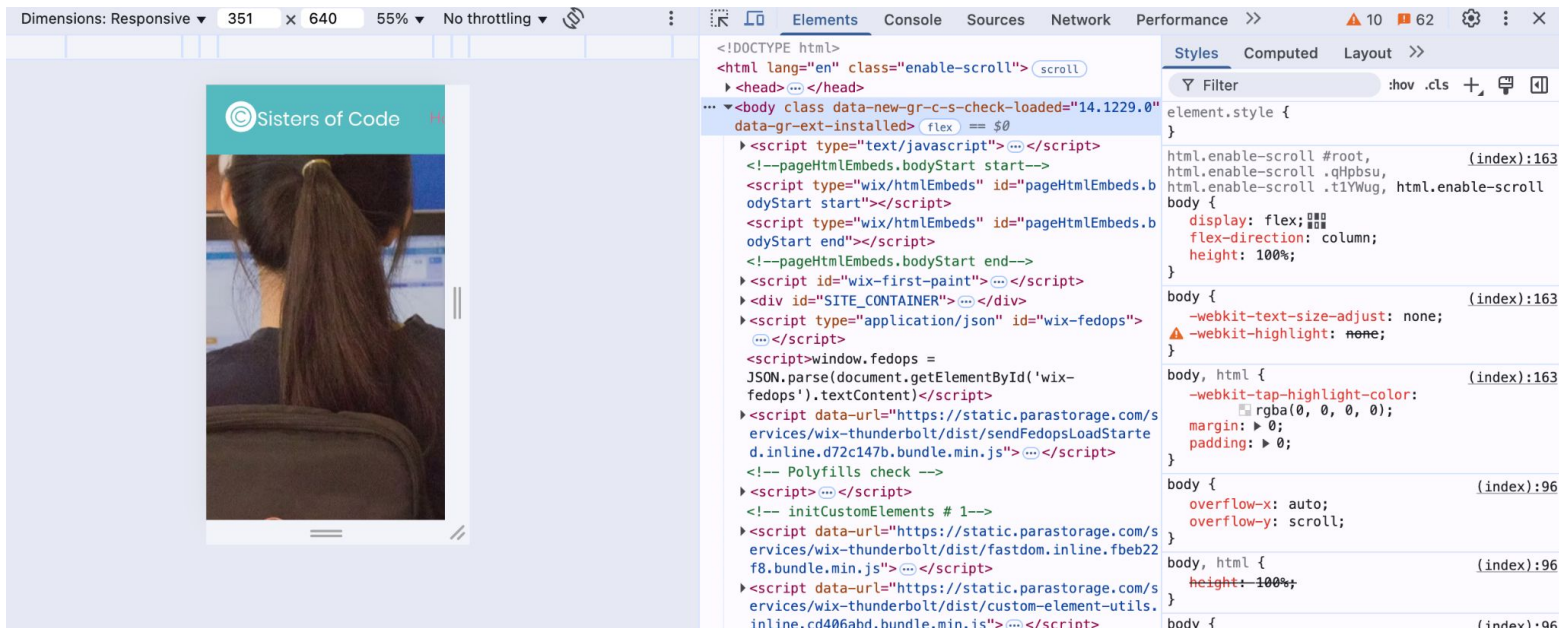
Frontend vs. Backend Development

- ❖ Frontend (Client-Side): Everything the user sees (HTML, CSS, JS).
- ❖ Backend (Server-Side): Manages data and server requests (PHP, Node.js, Python).



Google Chrome Developer Tool

Chrome DevTools is your secret weapon built right into the Chrome browser! It's a set of powerful tools that lets you peek under the hood of any website. You can see the code (HTML, CSS, JavaScript), check how fast things load, debug errors, and even play around with the styling in real-time. Think of it as a web developer's magnifying glass and control panel all in one!



Setting Up a Local Development Environment

IDE (Integrated Development Environment) is a software application that provides a comprehensive environment for software development, including code editing, debugging, and building, while a code editor is a tool primarily focused on writing and formatting code.

- VS Code
 - <https://code.visualstudio.com/download>

Version Control is a method for tracking and managing changes to files, especially code, over time, enabling developers to collaborate effectively and revert to previous versions if needed.

- Git: A system for tracking code changes.
 - <https://git-scm.com/downloads>
 - <https://desktop.github.com/download/>
- GitHub: A platform for storing and sharing projects.
 - <https://github.com>

Key takeaways

- ★ File management & browser basics
- ★ How the internet and websites work
- ★ Website fundamentals (domains, hosting, URLs)
- ★ Web development (HTML, CSS, JS)
- ★ Web hosting and website setup
- ★ Setting up a local development environment

After class activities

- Review today lessons again.
- Browse websites and practice defining Host, Domain, URL, Path, Port, etc. to get familiar.
- Use the Visual Studio Code to be familiar with tools and settings.
- Choose your favorite 3-5 websites and explore Google Chrome Developers tool to see how the websites look like on the backend
- Next week, we will learn Web UX UI Design



Thank you!

Practice makes perfect

